

ADVANCED SQL OF DATA ANALYTICS

QUESTION 1.

What is a Common Table Expression (CTE), and how does it improve SQL query readability?

> A Common Table Expression (CTE) is a temporary result set that you can reference within a SELECT, INSERT, UPDATE, or DELETE statement. It is defined using the WITH keyword.

How it improves readability:

- **Modularization:** It breaks down complex, nested queries into smaller, named blocks.
- **Sequential Logic:** You can read the query from top to bottom, similar to how you would read a script.
- **Recursion:** It allows for recursive queries (useful for hierarchical data like org charts) that are much cleaner than standard joins.

QUESTION 2.

Why are some views updatable while others are read-only? Explain with an example.

> A view is updatable if the database can map the change directly back to a single row in the underlying physical table. It becomes read-only if the view contains complex logic that makes the "source" of the data ambiguous.

- **Updatable View:** A simple `SELECT * FROM Products WHERE Category = 'Electronics'`. If you update a price here, the database knows exactly which row in the Products table to change.
- **Read-Only View:** A view using Aggregates (like SUM or AVG) or GROUP BY.

Example: A view showing total sales per category (`SELECT Category, SUM(Price) FROM Products GROUP BY Category`). If you try to update the "Total Price," the database doesn't know which individual product's price to change to reach that new sum.

QUESTION 3.

What advantages do stored procedures offer compared to writing raw SQL queries repeatedly?

- **Performance:** Stored procedures are compiled once and stored in the database, reducing parsing and execution time.
- **Reduced Network Traffic:** Instead of sending 50 lines of SQL code over the network, you send a single command: `EXEC GetMonthlyReport`.

- **Security:** You can give users permission to execute a procedure without giving them direct access to the underlying tables.
- **Maintainability:** If the logic changes, you only update it in one place (the procedure) rather than in every application that calls it.

QUESTION 4.

What is the purpose of triggers in a database? Mention one use case where a trigger is essential.

> A trigger is a special type of stored procedure that automatically "fires" or executes when a specific event (like INSERT, UPDATE, or DELETE) occurs in a table.

Essential Use Case: Audit Logging.

When a sensitive piece of data (like a user's salary or a product price) is updated, a trigger can automatically insert a record into an Audit_Log table with the old value, the new value, the timestamp, and the user who made the change. This ensures accountability that cannot be bypassed by the application.

QUESTION 5.

Explain the need for data modelling and normalization when designing a database.

- **Data Modeling** is the blueprint phase. It helps stakeholders understand how data entities relate to each other before any code is written, ensuring the system meets business requirements.
- **Normalization** is the process of organizing data to:

Reduce Redundancy: Prevents the same information from being stored in multiple places.

- **Ensure Data Integrity:** By reducing redundancy, you eliminate "Update Anomalies" (e.g., changing a supplier's phone number in one row but forgetting to change it in another).

Optimize Storage: Storing data efficiently saves space and can improve search performance when indexed correctly.